Modification of Dijkstra's Algorithm for Shortest Path Calculation

1. Problem Statement

Dijkstra's algorithm is used to find the shortest distances between vertices in a weighted graph. The program is modified to select a different source vertex and recalculate shortest distances, display the complete shortest path to a destination vertex, and update an edge weight before recalculating the shortest paths.

2. Objective

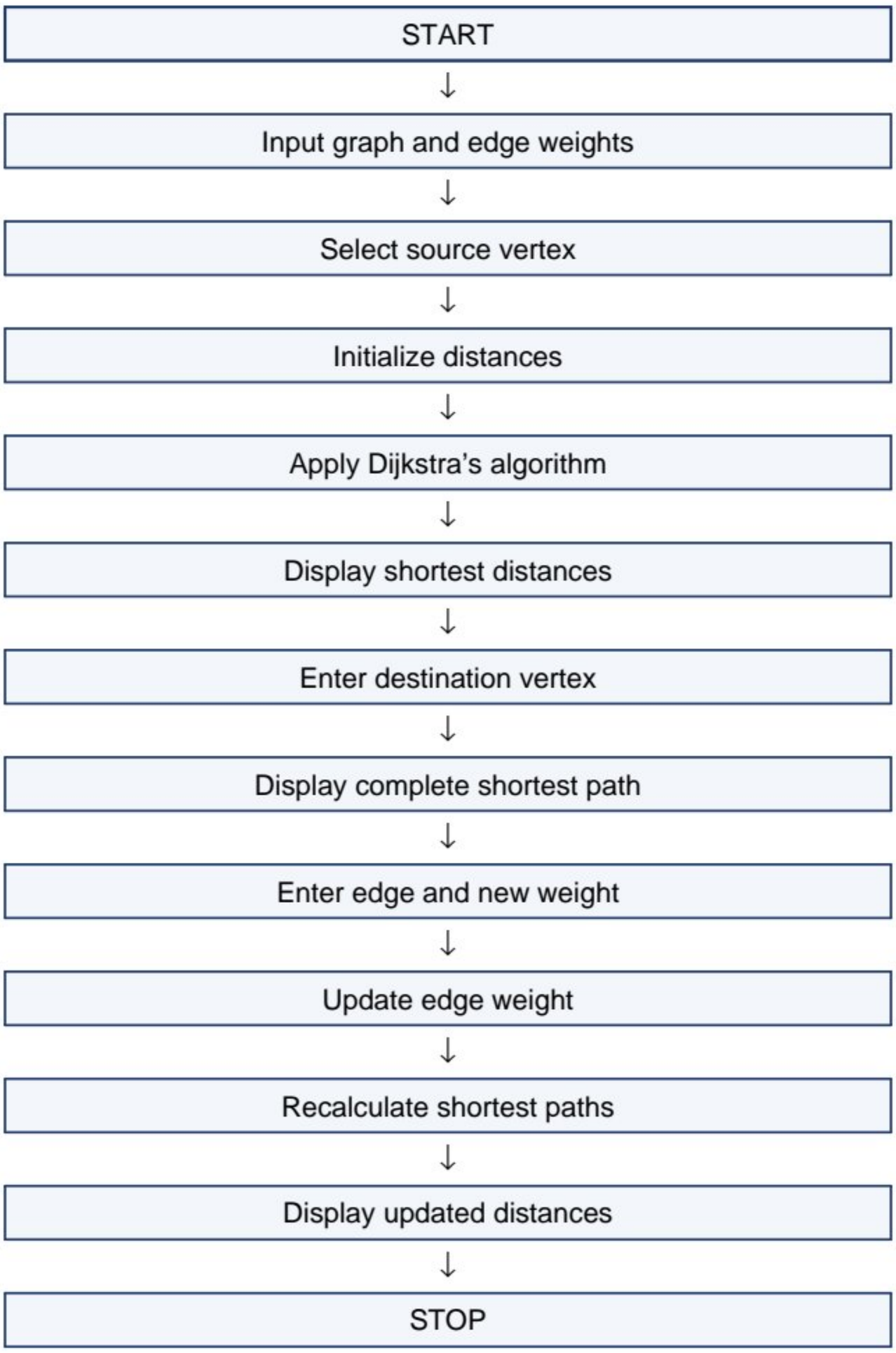
- To understand Dijkstra's shortest path algorithm.
- To calculate shortest distances from different source vertices.
- To display the complete shortest path to a destination vertex.
- To update an edge weight and recalculate shortest distances.

3. Algorithm

1. Start the program.
2. Read the number of vertices and the adjacency matrix.
3. Select the source vertex.
4. Initialize all distances to infinity and set the source distance to zero.
5. Apply Dijkstra's algorithm to calculate shortest distances.
6. Display the shortest distances from the selected source.
7. Enter a destination vertex and display its complete shortest path.
8. Enter an edge and its new weight.
9. Update the edge weight in the graph.
10. Run Dijkstra's algorithm again and display the updated shortest distances.
11. Stop the program.

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4. Flow Chart



5. Conclusion

The modified program uses Dijkstra's algorithm to calculate shortest distances from a selected source, display the complete shortest path to a destination, and recalculate the results after an edge weight is changed. These modifications make the program more flexible for analysing weighted graphs. The algorithm assumes non-negative edge weights.

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